



Presentation Title: Cellular Networks: Technologies like 4G (LTE) and 5G

Course Title: Wireless and Mobile Communication

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Presentation Overview

1. Introduction to Cellular Networks
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5. 4G vs 5G: Real-Life Impact
6. Why 5G is Better Than 4G
7. Benefits of 5G for Businesses
8. Challenges in 5G Implementation
9. Trends and the Future of Telecommunications

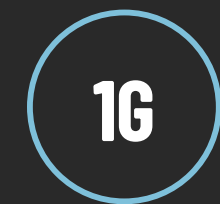
Introduction to Cellular Networks

Cellular networks are communication systems that divide geographic areas into small units called cells. Each cell is served by a base station (or cell tower) that connects mobile devices to the network using radio signals.

These networks enable wireless voice calls, text messaging, and internet access without the need for wired connections.

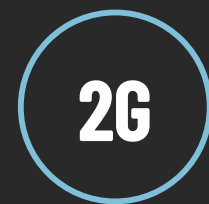


Evolution of Cellular Networks



1980s

A network limited to only voice call capabilities.



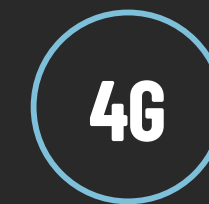
1990s

GSM, GPRS and EDGE advancements



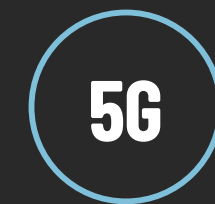
2000s

Allowed faster communication, web browsing and video streaming in smartphones



2010s

Greatly impacted the requirements set by IMT-Advanced standards.



Current

Significantly faster speeds with lower requirements to support IoT devices

What is 4G (LTE)?

4G, short for Fourth Generation, represents a major leap in mobile communication technology, focusing on high-speed internet and seamless data transfer.

It uses LTE (Long Term Evolution) — an advanced wireless standard that delivers faster and more stable connections than previous generations.

Introduced around 2009–2010.

Speed: Up to 100 Mbps download.

All-IP architecture for high efficiency.

Supports HD streaming, gaming, and VoLTE.



What is 5G Technology?

5G, or Fifth Generation Mobile Network, is the latest wireless technology that delivers ultra-fast internet speeds, extremely low latency, and supports massive device connectivity.

Introduced around 2019–2020, it can reach speeds up to 10 Gbps with latency of less than 1 millisecond, enabling real-time communication.

5G supports new technologies like smart cities, autonomous vehicles, IoT, and remote healthcare through its three key service types:

- **eMBB** (Enhanced Mobile Broadband)
- **URLLC** (Ultra-Reliable Low Latency Communication)
- **mMTC** (Massive Machine-Type Communication)

5G Technology:

Why It's Revolutionary:

5G enables smart cities, self-driving cars, telemedicine, and robotic automation.

It connects everything — people, devices, and machines — in real time.

Example:

A self-driving car communicates instantly with traffic lights and other vehicles to avoid accidents.

4G vs 5G: Real-Life Impact

Everyday Applications:

4G:

- High-speed streaming (YouTube, Netflix)
- Mobile banking and digital payments
- Smooth online gaming and social media

5G:

- Smart homes with connected devices
- Autonomous cars and drones
- Telemedicine and remote surgery
- Virtual reality (VR) and augmented reality (AR)
- Industrial automation and robotics

Economic & Social Impact:

- 4G built the digital economy — e-commerce, food delivery, and video streaming services.
- 5G will power the smart economy — automation, IoT, and artificial intelligence integration

Real-World Example:

 South Korea, China, and the USA are leading in 5G deployment.

Hospitals use 5G for real-time robotic surgeries, while factories use it for automated production.

Why 5G Is Better Than 4G

The next generation of wireless technology that will transform life significantly



5G technology guarantees a faster and smoother transmission of data, no matter how high the amount. Users can look forward to faster downloads and better support for more connected devices.

Before 5G Technology

- The pre-5G internet experience was slower and limited to certain spectrum types.
- Previous mobile network generations did not entirely support new services that rely heavily on communications.
- There were significantly more issues and limitations to availability and reliability.

With 5G Technology

- The pre-5G internet experience was slower and limited to certain spectrum types.
- Previous mobile network generations did not entirely support new services that rely heavily on communications.
- There were significantly more issues and limitations to availability and reliability.

Benefits of 5G for Businesses

5G technology can support operations to improve processes, produce better results and remain at the top of their game



NO NETWORK ISSUES

5G will bring a connection speed that's significantly faster than 4G, meaning more data transfer in less time, promising results without delay.



LOWER LATENCY

Lower latency will improve IoT devices' safety and functionality, allowing real-time viewing and guaranteeing zero lags in operations.



GREATER DATA CAPACITY

5G can support more devices and allow simultaneous data transfer because of its larger spectrum band.



Challenges in 5G Implementation

As exciting and appealing as 5G technology is, there are a few cons to it: its impact on battery life, high costs for rollout and imbalance in upload and download speeds.

- High infrastructure and deployment costs.
- Spectrum licensing and interference issues.
- Device compatibility and upgrade costs.
- Cybersecurity and privacy concerns.

Trends and the Future of Telecommunications

- 1 Artificial intelligence that keeps behind-the-scenes operations rolling faster will guarantee business optimization like never before.
- 2 Robotic Process Automation will significantly reduce error rates and boost customer service/operational efficiency.
- 3 Cloud technology will allow telecoms to operate with more efficiency as this guarantees scalability and cost-effectiveness, allowing them to adjust better to market demands.
- 4 The IoT or the Internet of Things is a technology that helps ensure minimal downtime for data centers and base stations. The telecom industry will continue to develop its services to improve IoT technology.
- 5 In-house software development is crucial in ramping up security and productivity. New software (with new tools and services that shift security efforts) will prevent any technological breach leading to systemwide failures.

Thank You